

These are notes I'm taking for Ben Hayes's Ergodic Theory class for Fall 2026.

The main texts are Petersen's *Ergodic Theory* and Walters's *An Introduction to Ergodic Theory*, as well as some other

General Motivation and Common Examples

The basic structure of ergodic theory is a standard probability space (X, μ) and, in its most basic form, a measurable map $T: X \rightarrow X$ satisfying $T_*\mu = \mu$. Recall that the pushforward measure is defined by

$$T_*\mu(A) = \mu(T^{-1}(A)).$$

We call T a measure-preserving transformation.

Generalizing this, if T is invertible, then we may take a group action $G \curvearrowright (X, \mu)$ by measure-preserving transformations. If T is not invertible, then we may consider a semigroup action $N \curvearrowright (X, \mu)$.

The main items we are interested in are analytical considerations, such as

- asymptotic structure
- connections to unitary representation theory
- isomorphism.

Ergodic theory appears in a number of fields, such as

- unitary representation theory
- analytic and geometric group theory
- functional analysis
- probability theory
- geometry
- analytic number theory.

Background

More aptly, this may be written as “background.” Most of these results are covered in a functional analysis class.

Definition: A measurable space (X, Σ) is called *standard Borel* if there is a separable and completely metrizable space (also known as a Polish space) K such that $(K, \mathcal{B}) \cong (X, \Sigma)$ are isomorphic as measurable spaces, where $\mathcal{B}$ denotes the Borel σ -algebra on K .

Remark: In the special case of standard probability spaces, we may take K to be compact. This emerges from the fact we are only concerned with Radon measures.

Example: The following are examples:

- finite or countably infinite sets;
- products of standard Borel spaces;
- $[0, 1] \cong S^1$;
- $(0, 1) \cong \mathbb{R} \cong [0, 1] \times \mathbb{Z}$;
- $\mathbb{R}^{\mathbb{N}}$, where we use the complete distance metric

$$d(t, s) = \sum_{j=1}^{\infty} 2^{-j} \frac{|t_j - s_j|}{1 + |t_j - s_j|}$$

to provide the desired isomorphism;

- the set $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ subset of the Polish space $\mathbb{R}$, so it is also a Polish space.

Theorem: If (X, Σ) is standard Borel and B is Borel, then $(B, \Sigma|_B)$ is standard Borel.

Theorem (Riesz Representation Theorem): Let K be a compact Hausdorff space, and $L: C(K) \rightarrow \mathbb{C}$ a continuous linear functional. Then, there is a unique complex-valued Borel measure such that

$$L(f) = \int f d\mu.$$

Furthermore, this measure is a Radon measure.

Remark: From the Riesz Representation Theorem, it can be shown that the following hold:

- L is positive if and only if μ is a positive measure;
- $L(1) = 1$ if and only if μ is a probability measure.

One of the fundamental constructions in ergodic theory, especially when we are concerned with actions by a countable discrete group on a standard probability space, is the projective limit of a (countable) family of standard probability spaces. This is a vital theorem.

Consider a system of standard probability spaces $(X_k, \Sigma_k)_{k=0}^\infty$ with connecting maps $\varphi_{k+1}: X_{k+1} \rightarrow X_k$ satisfying $(\varphi_{k+1})_* \mu_{k+1} = \mu_k$.

Then, the projective limit

$$\begin{aligned} X_\infty &= \varprojlim X_k \\ &= \left\{ (x_k)_k \in \prod_{k=1}^\infty X_k : \varphi_k(x_{k+1}) = x_k \right\} \end{aligned}$$

admits a family of projection maps $\varphi_{k,\infty}: X_\infty \rightarrow X_k$, where we have $\varphi_{k,\infty}((x_n)_n) = x_k$ such that $\varphi_k \circ \varphi_{k+1,\infty} = \varphi_{k,\infty}$.

$$\begin{array}{ccc} X_\infty & & \\ \varphi_{k+1,\infty} \downarrow & \searrow \varphi_{k,\infty} & \\ X_{k+1} & \xrightarrow{\varphi_k} & X_k \end{array}$$

The projective limit is universal in the sense that if we have a family of maps $f_{k+1}: Y \rightarrow X_{k+1}$ satisfying $\varphi_k \circ f_{k+1} = f_k$, then there is a unique map $f: Y \rightarrow X$ such that $\varphi_{k,\infty} \circ f = f_k$ for all k .

We define the σ -algebra on X_∞ to be

$$\Sigma_\infty = \sigma \left(\bigcup_{k=1}^\infty \{ \varphi_{k,\infty}^{-1}(E_k) : E_k \in \Sigma_k \} \right).$$

Then, the **Kolmogorov Extension Theorem** says the following.

Theorem (Kolmogorov Extension Theorem): There is a unique probability measure μ_∞ on $(X_\infty, \Sigma_\infty)$ satisfying

$$\mu_\infty(\varphi_{k,\infty}^{-1}(E)) = \mu_k(E)$$

for all $E \in \Sigma_k$.

We will give two proofs: one uses the Carathéodory criterion and is more explicit, while the other uses the Riesz Representation Theorem and is more slick.

Proof via the Riesz Representation Theorem. Since all the maps φ_k are Borel, there are zero-dimensional Polish topologies on each set X_k such that the φ_k are continuous; since the X_k are all probability spaces and

the measures μ_k are regular, it follows that each X_k is compact. In particular, we also have that the maps $\varphi_{k,\infty}$ are continuous.

Using Gelfand duality, it follows that for each X_k , there is an embedding $C(X_k) \rightarrow C(X_\infty)$ such that for any $f \in C(X_k)$, we obtain a continuous map $f \circ \varphi_{k,\infty}: X_\infty \rightarrow \mathbb{C}$. The set

$$A = \bigcup_{k \geq 1} C(X_k),$$

viewing each $C(X_k)$ as a subset of $C(X_\infty)$, is a unital $*$ -subalgebra that separates the points of X_∞ . We may define the linear functional $L: A \rightarrow \mathbb{C}$ given by taking

$$L|_{C(X_k)} = \int \cdot d\mu_k.$$

Notice that this is a positive linear functional satisfying $L(1) = 1$, so since L is continuous on the unital $*$ -subalgebra A , it follows that there is a unique extension to the closure $L_\infty: \overline{A} = C(X_\infty) \rightarrow \mathbb{C}$ satisfying $L_\infty(1) = 1$. By the Riesz Representation Theorem, there is a unique measure μ_∞ such that

$$L_\infty(f) = \int f d\mu_\infty$$

and satisfies

$$L_\infty|_{C(X_k)} = \int \cdot d\mu_k.$$

□

Proof via Carathéodory Criterion. Without loss of generality, we may assume that the X_k are compact and the φ_{k+1} are Borel. Consider the algebra

$$\mathcal{A} = \bigcup_{k \geq 1} \{ \varphi_{k,\infty}^{-1}(E) : E \in \Sigma_k \}.$$

Notice that this is the algebra of subsets that generates Σ_∞ . Furthermore, there is a unique set-map $\mu_\infty: \mathcal{A} \rightarrow [0, 1]$ taking $\mu_\infty(\varphi_{k,\infty}^{-1}(E)) = \mu_k(E)$ for all $E \in \Sigma_k$. Thus, it suffices to show that μ_∞ is a premeasure.

It suffices to show that for all decreasing sequences $(F_n)_n$ with $\mu_\infty(F_n) \geq \varepsilon > 0$, then the intersection of all the F_n is nonempty. This follows from the fact that if one has an ascending sequence $E_n \nearrow E$, then $(E \setminus E_n) \searrow \emptyset$, so if one takes the Carathéodory criterion and takes the contrapositive of this reversal, we obtain the desired result.

Let

$$F_k = \varphi_{n_k,\infty}^{-1}(E_{n_k}).$$

If n_k does not diverge to ∞ , then there exists some subsequence n_{k_j} that is constant equal to n , and F_{k_j} all have $\varphi_{n,\infty}(F_{k_j}) = E_{n_{k_j}}$. Since μ_n is a measure, we are done in this case, following from the fact that all the F_n past F_{k_j} lie in X_n . □

The most important case is when the projective limit is the infinite product measure. If (X_n, μ_n) is a countable family of standard probability spaces, then the product measure $\bigotimes_{n=1}^{\infty} \mu_n$ is a probability measure.

Canonical Examples

Example (Bernoulli Actions): Let (X, μ) be a standard probability space, and let G be any countable group. Then, there is a unique probability measure on the product space $(X, \mu)^G$, and we have that $G \curvearrowright (X, \mu)^G$ in a probability measure-preserving manner by translation — that is, $h \cdot (x_g)_{g \in G} = (x_{h^{-1}g})_{g \in G}$. This follows from the fact that G is probability measure-preserving on all the Borel cylinder sets

$$\bigcup_{\substack{F \subseteq G \\ F \text{ finite}}} \{ \pi_F^{-1}(E) : E \subseteq X^F \},$$

so we may apply the monotone class lemma (that no one remembers from measure theory) to see that it applies for all the Borel sets of $(X, \mu)^G$.

Example: Suppose G is a compact Hausdorff topological group. Then, there is a unique (regular) Borel probability measure μ , known as the Haar (probability) measure that is invariant under left translation. If $\pi: \Gamma \rightarrow G$ is a group homomorphism of a discrete group Γ on G , then the action $\gamma \cdot g = \pi(\gamma)g$ yields a probability measure-preserving action of Γ on G .

For instance, this yields actions $\mathbb{Z} \curvearrowright S^1$ given by rotations — that is, $e^{2\pi i \theta} \mapsto e^{2\pi i(\theta + \alpha)}$.

In general, if G is a compact group, and $\phi: \Gamma \rightarrow \text{Aut}(G)$ is an action of a discrete group Γ that preserves the Haar measure, we call such an action an *algebraic action*.

Example: If M is a finite volume Riemannian manifold with volume form vol , and $\phi \in \text{Diff}(M)$ is a diffeomorphism satisfying $\det(\phi'(p)^* \phi'(p)) = 1$ for all $p \in M$, then the pushforward measure satisfies $\phi_* \text{vol} = \text{vol}$.

Group actions $\Gamma \rightarrow \text{Diff}(M)$ appear in geodesic and [horocycle flow](#).

The Ergodic Theorems

We will now discuss the mean and pointwise ergodic theorems.

Mean Ergodic Theorem

Proposition: Let X be a Banach space. Let $(T_n)_n \in B(X)$ be a uniformly operator-norm bounded. Then, for all $S \in B(X)$, the sets

$$\begin{aligned} V &= \{x \in X : T_n x \text{ converges}\} \\ W &= \{x \in X : \|T_n x - Sx\| \rightarrow 0\} \end{aligned}$$

are norm-closed subspaces of X .

Theorem (Mean Ergodic Theorem): Let H be a Hilbert space, and let $T \in B(H)$ be such that $\|T\| \leq 1$. Define

$$V = \{\xi \in H : T\xi = \xi\},$$

and let $P = \text{proj}_V$. Then,

$$\left(\frac{1}{n} \sum_{j=0}^{n-1} T^j \right) \xrightarrow{\text{SOT}} P.$$

Proof. Let $K_1 = \text{im}(1 - T)$ and $K_2 = \ker(1 - T)$. We claim that K_1 and K_2 are orthogonal to each other and satisfy $\overline{K_1} + K_2 = H$. For the first point, we have for all $v \in K_1$ and $w \in K_2$, we write $v = (1 - T)\xi$, and have

$$\langle (1 - T)\xi, w \rangle = \langle \xi, w \rangle - \langle \xi, T^* w \rangle$$

$$\begin{aligned}
\|T^*w - w\|^2 &= \|T^*w\|^2 - 2\operatorname{Re}(\langle w, Tw \rangle) + \|w\|^2 \\
&= \|T^*w\|^2 - \|w\|^2 \\
&\leq 0,
\end{aligned}$$

where we note that $\|T^*\| = \|T\|$. Therefore, $T^*w = w$, so we have that $K_2 \subseteq K_1^\perp$.

Now, suppose $\xi \in K_1^\perp$. Then,

$$\|(1 - T)\xi\|^2 = \|T\xi\|^2 - 2\operatorname{Re}(\langle T\xi, \xi \rangle) + \|\xi\|^2,$$

with

$$\begin{aligned}
\langle T\xi, \xi \rangle &= \langle (T - 1)\xi, \xi \rangle + \|\xi\|^2 \\
&= \|\xi\|^2
\end{aligned}$$

since $\xi \in K_1^\perp$. Therefore,

$$\begin{aligned}
\|(1 - T)\xi\|^2 &\leq \|\xi\|^2 - \|\xi\|^2 \\
&= 0
\end{aligned}$$

since $\|T\| \leq 1$. Therefore, $K_2 = K_1^\perp$.

Therefore, it suffices to show that

$$\frac{1}{n} \sum_{j=0}^{n-1} T^j \xi \rightarrow P\xi$$

for all $\xi \in K_1 \cup K_2$. For each $\xi \in K_2$, we obtain the desired result since both sides are equal to ξ .

For each $\xi \in K_1$, we have that $P\xi = 0$. If $\xi = (1 - T)\eta$, then

$$\frac{1}{n} \sum_{j=0}^{n-1} T^j \xi = \frac{1}{n} (1 - T^n) \eta,$$

so that

$$\begin{aligned}
\left\| \frac{1}{n} \sum_{j=0}^{n-1} T^j \xi \right\| &\leq \frac{2}{n} \|\eta\| \\
&\xrightarrow{n \rightarrow \infty} 0.
\end{aligned}$$

□

Pointwise Ergodic Theorem

Now, consider a probability space (X, μ) and a measure-preserving map $T: X \rightarrow X$. Then, the operator $V_T \in B(L^2(X, \mu))$ is given by $V_T f = f \circ T$ is an isometry.

If $f \in L^\infty(X, \mu)$, and $P = \operatorname{proj}(\operatorname{fix}(V_T))$, then we have $\|Pf\|_{L^\infty} \leq \|f\|_{L^\infty}$. This follows from the fact that we may find a subsequence such that

$$\frac{1}{n_k} \sum_{j=0}^{n_k-1} f \circ T^j \rightarrow Pf$$

pointwise almost everywhere. Therefore, we have

$$\begin{aligned}\|Pf\|_{L^\infty} &\leq \liminf_{k \rightarrow \infty} \frac{1}{n_k} \left\| \sum_{j=0}^{n_k-1} f \circ T^j \right\|_{L^\infty} \\ &\leq \|f\|_{L^\infty}.\end{aligned}$$

A similar result holds on L^p for general p . We discuss this below.

Corollary: Suppose (X, μ) is a probability space, and $T: X \rightarrow X$ is a probability measure-preserving action. Then, for all $f \in L^p(X, \mu)$ with $1 \leq p < \infty$, there exists $\tilde{f} \in L^p(X, \mu)$ such that

$$\left\| \tilde{f} - \frac{1}{n} \sum_{j=0}^{n-1} f \circ T^j \right\|_{L^p} \xrightarrow{n \rightarrow \infty} 0.$$

The element $\tilde{f}$ satisfies:

- (i) $\tilde{f} \circ T = \tilde{f}$ almost everywhere;
- (ii) $\int \tilde{f} g d\mu = \int f g d\mu$ for all $g \in L^q(X, \mu)$ satisfying $g \circ T = g$ almost everywhere, where q is the conjugate exponent for p ;
- (iii) $\|\tilde{f}\|_{L^p} \leq \|f\|_{L^p}$.

Proof. In the following proof, we will still be concerned with the operator P defined as

$$\begin{aligned}P &= \text{proj}(\text{fix}(V_T)) \\ &= \text{proj}\left(\left\{f \in L^2(X, \mu) : V_T f = f\right\}\right).\end{aligned}$$

- (i) Let $V_T \in B(L^p(X, \mu))$ be defined by $V_T f = f \circ T$. Then,

$$\left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j \right\| \leq 1.$$

We have that

$$W = \left\{ f \in L^p(X, \mu) : \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f \text{ converges} \right\}$$

is a norm-closed subspace. It suffices to show that $L^\infty(X, \mu) \subseteq W$, as then the L^p -closure of $L^\infty(X, \mu)$ is contained in W , but that is equal to $L^p(X, \mu)$.

Fix $f \in L^\infty(X, \mu)$.

If $1 \leq p \leq 2$, we have that

$$\begin{aligned}\left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^p} &\leq \left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^2} \\ &\rightarrow 0.\end{aligned}$$

If $p \geq 2$, then

$$\begin{aligned}\left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^p}^p &\leq 2\|f\|_{L^\infty}^{p-2} \left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^2}^2 \\ &\rightarrow 0.\end{aligned}$$

(ii) We have that $(fg) \circ T^j = (f \circ T^j)g$, so that

$$\left\| \tilde{f}g - \frac{1}{n} \sum_{j=0}^{n-1} (f \circ T^j)g \right\|_{L^1} \leq \|g\|_q \left\| \tilde{f} - \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f \right\|_{L^p}.$$

In particular, we may write $\tilde{f}g$ using the asymptotic form for $\tilde{f}$. Therefore, we have

$$\begin{aligned} \int \tilde{f}g \, d\mu &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \int f(T^j(x))g(x) \, d\mu(x) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \int (fg)(T^j(x)) \, d\mu(x) \\ &= \int fg \, d\mu. \end{aligned}$$

(iii) We have that

$$\begin{aligned} \|\tilde{f}\|_{L^p} &= \lim_{n \rightarrow \infty} \frac{1}{n} \left\| \sum_{j=0}^{n-1} f \circ T^j \right\|_{L^p} \\ &\leq \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \|\tilde{f} \circ T^j\|_{L^p} \\ &= \|f\|_{L^p}. \end{aligned}$$

□

The next result we will be concerned with is the Pointwise Ergodic Theorem. To prove it, we must prove a version of this theorem known as the Maximal Ergodic Theorem. This concerns a maximal function defined by

$$f^*(x) = \sup_n \frac{1}{n} \sum_{j=0}^{n-1} f \circ T^j(x).$$

Theorem: If $f \in L^1(X, \mu)$, then

$$\int_{\{f^* > 0\}} f \, d\mu \geq 0.$$

Proof. The set on which f^* is strictly greater than zero is the disjoint union of the sets

$$\begin{aligned} B_1 &= \{x : f(x) \geq 0\} \\ &\vdots \\ B_n &= \{x : f(x) \leq 0, f(x) + f \circ T(x) \leq 0, \dots, f(x) + \dots + f \circ T^{n-2}(x) \leq 0, f(x) + \dots + f \circ T^{n-1}(x) > 0\}. \end{aligned}$$

If we are able to show that

$$\int_{B_1 \cup \dots \cup B_n} f \, d\mu \geq 0$$

holds, then dominated convergence yields the desired result.

In order to obtain this, we will break up the set $B_1 \cup \dots \cup B_n$ into a family of disjoint pieces on which the

integral of f is strictly positive. We will view them as towers of the form $B'_k \cup T(B'_k) \cup \dots \cup T^{k-1}(B'_k)$.

First, we observe that $T^k(B_n) \subseteq B_1 \cup \dots \cup B_{n-k}$ for each $k = 1, \dots, n-1$, since if $x \in B_n$, then we have

$$0 \geq f(x) + f \circ T(x) + \dots + f \circ T^{k-1}(x)$$

while

$$0 < f(x) + f \circ T(x) + \dots + f \circ T^{k-1}(x) + f \circ T^k(x) + \dots + f \circ T^{n-1}(x).$$

In particular, this means that

$$\begin{aligned} 0 &< f \circ T^k(x) + \dots + f \circ T^{n-1}(x) \\ &= f(T^k(x)) + f \circ T(T^k(x)) + \dots + f \circ T^{n-k-1}(T^k(x)). \end{aligned}$$

Next, we note that the sets $B_n, T(B_n), \dots, T^{n-1}(B_n)$ are pairwise disjoint. This follows from the fact that if $T^i(B_n) \cap T^j(B_n) \neq \emptyset$ for some $i < j$, then $B_n \cap T^{j-i}(B_n) \neq \emptyset$, but we have already established the family of sets B_i are disjoint, while $T^{j-i}(B_n) \subseteq B_1 \cup \dots \cup B_{n-(j-i)}$.

We will define the decomposition of $B_1 \cup \dots \cup B_n$ as follows. Take

$$\begin{aligned} B'_n &= B_n \\ C_n &= B'_n \cup T(B'_n) \cup \dots \cup T^{n-1}(B'_n) \\ B'_{n-1} &= B_{n-1} \setminus C_n \\ C_{n-1} &= B'_{n-1} \cup T(B'_{n-1}) \cup \dots \cup T^{n-2}(B'_{n-1}) \\ &\vdots \\ B'_1 &= B_1 \setminus (C_2 \cup \dots \cup C_n) \\ C_1 &= B'_1. \end{aligned}$$

The picture one must have in mind is viewing each C_k as a “column” in a table with disjoint “levels” given by $T^j(B'_k)$ for each $0 \leq j \leq k-1$. Each “base” B'_k is disjoint □